

SUMMARY OF RESEARCH PROJECTS UNTIL ABOUT 2009

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I give a brief description of the projects I carried out until roughly 2009, putting them into context. The projects are grouped by topic as follows:

- (1) Reduction of submanifolds
- (2) Coisotropic embedding problems
- (3) Prequantization and Lie groupoids
- (4) Averaging of submanifolds

I follow roughly a chronological order within each section.

1. REDUCTION OF SUBMANIFOLDS

We use the term *reduction* to denote the following procedure: we consider a manifold M endowed with a geometric (for instance symplectic, Poisson,...) structure, a suitable submanifold with a natural foliation, and endow the leaf space of the submanifold with a geometric (symplectic, Poisson,...) structure. Standard examples are Marsden-Weinstein reduction in symplectic geometry and coisotropic reduction.

In this section I describe projects involving reduction in the frameworks of Jacobi-contact geometry (§1.1), Dirac geometry (§1.2), Poisson geometry (§1.3) and generalized geometry (§1.4).

1.1. Reduction of Jacobi manifolds via contact groupoids ([ZZ06]). Willett [Wil02] and Albert [Alb89] independently developed reduction procedures for contact manifolds: given a Hamiltonian action of a Lie group G on a contact manifold M with moment map $J : M \rightarrow \mathfrak{g}^*$ (the dual of the Lie algebra of G), they showed that the quotient of suitable preimages of J by certain subgroups of G are again contact manifolds. However these ways to perform reduction are not as natural as their counterpart in symplectic geometry (Marsden-Weinstein reduction). In the joint work [ZZ06] with Zhu we perform contact reduction using so-called contact groupoids instead of a group G . This generalizes Willett's and Albert's results, puts contact reduction into a more natural framework and is also suitable for computations.

Recall that a contact manifold (i.e. manifold equipped with a 1-form θ satisfying a non-degeneracy condition) is a special type of so-called *Jacobi manifold*. There are global objects associated to Jacobi manifolds, called contact groupoids, which “desingularize” the singular foliations with which Jacobi manifolds are equipped. Lie groupoids are generalizations of groups and are suitable to describe “global” behaviors; a *contact groupoid* is just a Lie groupoid endowed with a compatible contact 1-form. Let (M, θ) be a contact manifold, X a Jacobi manifold, Γ the contact groupoid associated to X . The main result of [ZZ06] is that a very explicit starting data, namely

a (complete) map $J : M \rightarrow X$ preserving the Jacobi structures, naturally induces a (contact) groupoid action of Γ on M , that the quotient M/Γ has an induced (conformal) Jacobi structure, and its leaves are precisely given by the quotients $J^{-1}(x)/\Gamma_x$ (where $x \in X$ and $\Gamma_x \subset \Gamma$ is suitable Lie group). The contact form on a contact leaf is induced (up to a conformal factor) by the contact one-form θ on M .

1.2. Stretching of Dirac structures ([CFZ10]). A *Dirac structure* over a manifold M is given by a subbundle L of $(TM \oplus T^*M, \langle \cdot, \cdot \rangle, [\cdot, \cdot]_H)$ which is maximal isotropic w.r.t. the natural symmetric pairing $\langle \cdot, \cdot \rangle$ and closed w.r.t. the H -twisted Courant bracket $[\cdot, \cdot]_H$, where H is a fixed closed 3-form. When $H = 0$ and L projects surjectively onto T^*M (resp. TM), then L is necessarily the graph of a Poisson bivector (resp. of a closed 2-form). Viewing Poisson and presymplectic manifolds as Dirac structures has an advantage: one can always push forward and pull back Dirac structures (at least pointwise) along maps, indeed also along Lagrangian relations.

In [CFZ10], with Calvo and Falceto, we produce new Dirac structures with the following recipe: start with a Dirac structure (M, L) , choose an isotropic subbundle $S \subset TM \oplus T^*M$, and define $L^S := (L \cap S^\perp) + S$. When it is smooth and closed under the Courant bracket, this is again a Dirac structure, the “stretching of L along S ”. An interesting example is when L is the graph of a Poisson bivector, because choosing $S \subset T^*M$ suitably one recovers the so-called Dirac bracket, used by Dirac in his theory of constraints. The stretching construction can be interpreted in terms of reduction when suitable assumptions are satisfied.

1.3. An extension of the Marsden-Ratiu reduction in Poisson geometry ([CZ13][CZ09a][FZ08]). The Marsden-Ratiu reduction theorem [MR86] dates back to 1986, and provides conditions on a submanifold C of a Poisson manifold (M, π) so that, extending certain functions on C to annihilate a fixed subbundle $E \subset TM|_C$, the quotient $C/(E \cap TC)$ inherits a Poisson structure. In the joint work [CZ09b] with Falceto we argue that the assumptions of the original Marsden-Ratiu theorem are too restrictive. We provide a reduction statement that works with much more general assumptions, and show that these are met in several cases of interest, such as slices transverse to zero level sets of symplectic moment maps. Incidentally, the Poisson structure on the quotient $C/(E \cap TC)$ corresponds to the Dirac structure on C obtained pulling back to C the stretching of $graph(\pi)$ along E .

A Poisson manifold M can be equivalently described by a self-commuting degree 2 function S on the graded manifold $T^*[1]M$. A choice of submanifold in $T^*[1]M$ is equivalent to a choice of a subbundle E over a submanifold $C \subset M$. In [CZ13][CZ09a] with Cattaneo we show that performing pre-symplectic reduction in stages at the graded level one not only recovers the Marsden-Ratiu reduction theorem, but also improves slightly the extensions made in [FZ08] by Falceto and the author.

1.4. Reduction of branes in generalized complex geometry ([JRZ12][Zam08b]). Recall that an *Courant algebroid* is a vector bundle $E \rightarrow M$ endowed with a symmetric non-degenerate pairing on the fibers, a bracket on its space of sections, and a morphism $\pi : E \rightarrow TM$ satisfying compatibility conditions. So-called *exact* Courant algebroids are all isomorphic to $(TM \oplus T^*M, \langle \cdot, \cdot \rangle, [\cdot, \cdot]_H, \pi = pr_{TM})$ that appeared

in §1.2 for some H , but the isomorphism depends on an arbitrary choice, so that it is preferable to work with an abstract exact Courant algebroid E . In §1.2 we defined the notion of Dirac structure $L \subset E$, and here we recall Hitchin's notion of *generalized complex structure* [Gua11]: it is an endomorphism $\mathcal{J} : E \rightarrow E$ preserving the symmetric pairing $\langle \cdot, \cdot \rangle$, squaring to -1 and satisfying an integrability condition. Generalized complex structures allow for the first time to threat on the same footing symplectic and complex structures, and became an intense object of study for physicists because they might be the right framework for mirror-symmetry and might relate the A and B -model in string theory. The symmetries of our exact Courant algebroid E of course also act on $\mathcal{J}$, and the only geometric feature of $\mathcal{J}$ which is invariant and intrinsic is the Poisson structure Π that it induces on M .

Bursztyn-Cavalcanti-Gualtieri [BCG07] carry out the reduction of exact Courant algebroids, Dirac structures and generalized complex structures in the presence of a group action, generating in particular non-trivial examples of generalized complex structures. In [Zam08b] I carry out the reductions of these three structures assuming that the symmetries arise not from a group action but rather from an isotropic subbundle of E with base some submanifold C of M . This is interesting in its own right, it sheds light on the role played by the so-called Ševera class of E (an element of $H^3(M, \mathbb{R})$), and can be used to produce new examples of generalized complex structures (a “dream” would be to give an answer by this method to the long-standing open question of whether there exists a complex structure on S^6). However the main point of [Zam08b] is to show that a natural class of submanifolds, called *branes* in the physics literature and *generalized complex submanifolds* by Gualtieri [Gua11], are coisotropic (w.r.t. the Poisson structure Π), and that their quotients $\underline{C}$ are generalized complex manifold again. In addition $\underline{C}$ is a “space-filling” brane for the reduced structure. A remarkable fact is that the above reduction works even if one drops the integrability condition on the brane; hence we hope to construct new examples of generalized complex structures with this procedure.

One of the reduction procedures developed in [Zam08b] is the reduction by distributions of Dirac structures. In [JRZ12] with Jotz and Ratiu we provide an alternative, more geometric proof of this procedure, and further we present applications to reduction under group actions which are proper but not necessarily free.

2. COISOTROPIC EMBEDDING PROBLEMS

We investigate the geometry of coisotropic submanifolds in symplectic manifolds (§2.1) and Poisson manifolds (§2.2). Further we use the geometry of coisotropic subgroups of Poisson-Lie groups to make a statement about Lie bialgebras (§2.3).

Recall that a *coisotropic* submanifold C of Poisson manifold (P, π) is one such that $\sharp N^*C \subset TC$, where $N^*C \subset T^*P|_C$ is the annihilator of TC and $\sharp : T^*P \rightarrow TP$ is just the contraction with the bivector π . The (usually singular) distribution $\sharp N^*C$, called characteristic distribution, is always integrable and the leaf space is a Poisson manifold whenever it is smooth. For more details see [Zam11b], where I present a survey on submanifolds of Poisson manifolds.

2.1. Deformations of coisotropic submanifolds ([Zam08a]). A coisotropic submanifold C of a symplectic manifold (M, ω) is one where the symplectic orthogonal TC^ω is contained in TC . Being a coisotropic submanifold is a quite rigid condition (except in special cases such as the lagrangian one); I could establish [Zam08a] that, notwithstanding this rigidity, the topological type of the foliation integrating TM^ω is *not* invariant under perturbations of the coisotropic submanifold. Furthermore, I could show that the space of coisotropic submanifolds nearby a given one N does not have a (Frechet) manifold structure. The topic of this work is closely related to the remarkable work [OP05] of Oh and Park that was completed at the same time. There it is shown that the deformation problem is governed by the structures of a L_∞ -algebra on the space of sections of $\wedge^\bullet NC$ (where NC is the normal bundle to C), and one of my examples from [Zam08a] is used to explain how the L_∞ -algebra degenerates in the presence of a transverse foliation. This L_∞ structure is attached to coisotropic submanifolds of Poisson manifolds too [CF07], and is the same that appears in §2.2 below.

2.2. Coisotropic embeddings in Poisson manifolds ([CZ07a][CZ09b]). Coisotropic submanifolds of Poisson manifolds (see §1) have been considered because they represent convenient boundary conditions (“branes”) in various sigma models. However more general kinds of submanifolds C of a Poisson manifold P , satisfying merely a constant rank condition and called *pre-Poisson submanifolds*, turn out to be suitable choices for branes [CF04]. The space of invariant functions on a pre-Poisson submanifold is a Poisson algebra, however one can usually not apply Kontsevich’s theorem [Kon03] to determine whether it admits a deformation quantization.

In the joint work [CZ09b] with Cattaneo we show that a pre-Poisson submanifold can be embedded coisotropically in a so-called Poisson-Dirac submanifold $\tilde{P}$ of P (and that $\tilde{P}$ is essentially unique). We deduce a statement about deformation quantization and the existence of an L_∞ structure associated to every pre-Poisson submanifold. In [CZ07a] we give some examples of pre-Poisson submanifolds of duals of Lie algebras and study further properties.

The second part of [CZ09b] deals with a different embedding problem: given an abstract Dirac manifold (M, L) , we show that it can be embedded coisotropically into some Poisson manifold if and only iff the characteristic distribution $TM \cap L$ has constant rank, generalizing a well-know theorem of Gotay in the symplectic setting. However the problem of proving the uniqueness of the Poisson manifold, even after imposing dimensional restrictions to it, is still open.

2.3. A construction of coisotropic subalgebras of Lie bialgebras ([Zam11a]). The starting point is given by considering maps between Poisson manifolds which are not Poisson morphisms, but whose graphs are pre-Poisson submanifolds (see §2.2). A natural example is the left translation on a Poisson-Lie group G . Analyzing further this example leads to a way to produce so-called coisotropic subalgebras of the Lie bialgebra $\mathfrak{g}$ of G , i.e. Lie subalgebras of $\mathfrak{g}$ whose annihilators are Lie subalgebras of $\mathfrak{g}^*$. Hence we are able to make algebraic statements on the bialgebra $\mathfrak{g}$ using the geometry of the underlying Lie-Poisson group G . Our procedure is systematic

and computationally friendly, leading to very explicit results for instance when $\mathfrak{g}$ corresponds to a simple complex Lie algebra. This is interesting as there seems to be no systematic way to produce coisotropic subalgebras; on the other hand our examples are always even dimensional, so they surely do not encompass all coisotropic subalgebras.

3. PREQUANTIZATION AND LIE GROUPOIDS

In this section we discuss projects concerning prequantization: prequantization of Dirac manifolds (§3.1), a study of the groupoids associated to the prequantization spaces (§3.2).

3.1. Geometric structures on prequantization spaces ([WZ05]). Geometric quantization is a geometric construction, developed by Kostant and Souriau in the 70's, to quantize a symplectic manifold (P, ω) , in the sense that it produces a suitable representation for the Poisson algebra $C^\infty(P)$. This can be carried out only when ω represents an integer cohomology class, and its first step (*prequantization*) consists of constructing a suitable circle bundle $\pi : Q \rightarrow P$ with connection (which turns out to be a contact 1-form) whose curvature is ω . Geometric quantization turned out to be useful in representation theory and provides interesting geometry. Poisson and more generally Dirac manifolds (see §1.2) also carry a Poisson algebra structure. In the work [WZ05] with Weinstein we clarified the geometric structures with which the prequantizing circle bundles Q of symplectic, Poisson and Dirac manifolds P are endowed. The notion of prequantizability used there was first formulated in [Vai94] and implies the one of [CZ07b] and [BCZ06]; in some respects the notion of prequantizability we used is not as well-behaved, but it presents non-trivial and interesting geometric features. We have the following diagram, where in the top row we characterize the geometric structure on the prequantization circle bundles Q of the prequantizable manifolds P beneath it:

$$\begin{array}{ccccc}
 \{Q \text{ contact}\} & \subset & \{Q \text{ Jacobi}\} & \subset & \{Q \text{ Jacobi} - \text{Dirac}\} \\
 \downarrow \pi & & \downarrow \pi & & \downarrow \pi \\
 \{P \text{ symplectic}\} & \subset & \{P \text{ Poisson}\} & \subset & \{P \text{ Dirac}\}
 \end{array}$$

3.2. Groupoids of prequantization spaces ([ZZ07]). The joint work [ZZ07] with Zhu considers the geometric relation between the Lie groupoids associated to the spaces considered in [WZ05] (see §3.1). If a Dirac (or, more restrictively, just a symplectic or Poisson) manifold P is prequantizable in the sense of [Vai94] then its symplectic groupoid $\Gamma_s(P)$ is automatically prequantizable, and it is known that one can put a unique structure of contact groupoid on the corresponding circle bundle $\Gamma_c(P)$. It is natural to wonder about the relation between the prequantization Q of P and the prequantization of the symplectic groupoid $\Gamma_s(P)$. Diagrammatically we

are in the following situation, where superscripts indicate dimensions:

$$\begin{array}{ccccc}
 \Gamma_c(Q)^{2n+3} & & \Gamma_c(P)^{2n+1} & \longrightarrow & \Gamma_s(P)^{2n} \\
 \Downarrow & & \Downarrow & \swarrow & \nearrow \\
 Q^{n+1} & \xrightarrow{\pi} & P^n & &
 \end{array}$$

We show that the S^1 action on Q lifts to $\Gamma_c(Q)$, the (pre-)contact groupoid of Q , whose Albert-Willett reduction (see §1.1) is exactly $\Gamma_c(P)$. It is interesting that both the (pre-)contact and groupoid structure descend to the quotient $\Gamma_c(P)$; the proof is a combination of finite dimensional techniques and constructions in certain path spaces, closely related to the Poisson-Sigma model. This geometric picture could prove to be useful in Weinstein’s quantization program via symplectic groupoids.

4. AVERAGING OF SUBMANIFOLDS

In this last section I describe briefly work that I carried out as part of my Ph.D. thesis.

The starting point is a purely Riemannian-geometric result of Weinstein’s [Wei83] in 1999, which shows the following: given a family of (nicely embedded) submanifolds $\{N_g\}$ of a Riemannian manifold M , there is a construction that produces a “center of mass” submanifold $\bar{N}$ (the “average” of the N_g ’s), and this construction is equivariant with respect to isometries of M . The construction is canonical, i.e. it does not involve arbitrary choices, and has applications for example to compact group actions.

I adapt this statement to the setting of symplectic geometry (§4.1) and contact geometry (§4.2).

4.1. Averaging of isotropic submanifolds ([Zam06]). In [Zam06] I specialized the above averaging procedure to the setting of symplectic geometry: starting with a family of isotropic submanifolds, I could produce canonically an isotropic average. The idea of the proof is to take Weinstein’s average - which will usually not be isotropic - and deform it canonically to an isotropic submanifold using Moser’s path method. The techniques used are all finite dimensional, but the result can be seen as an infinite dimensional statement (“In the space of isotropic submanifolds of M one can determine the center of mass of the family of points N_g ”).

4.2. Averaging of legendrian submanifolds ([Zam05]). In [Zam05] I proved a theorem analog to the one in [Zam06] for legendrian submanifolds of a (not necessarily co-orientable) contact manifold M by passing to the symplectization of M , and obtaining in this way better estimates than if I had used directly Moser’s path method on M .

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